

UNIT 4: E-Business Systems, E-Business Models & E-Commerce

This guide covers all key topics from Unit 4. Use it to review definitions, concepts, models, and distinctions that are likely to appear in your final exam. Each section is summarized for quick revision.

1.1 Key Topics to Master

The following are the core areas you must thoroughly understand before your exam:

- Definitions: e-business, e-commerce, e-marketing, m-commerce
- E-business types: B2B, B2C, C2C, C2B, B2G, G2B, G2C, G2G, B2E
- Pure Play vs. Brick and Click business models
- E-business security: concerns, measures, and solutions
- E-marketing: features, advantages, tools, and e-branding
- E-commerce: advantages, disadvantages, threats, EDI & EFT
- Key differences between e-commerce and e-business
- Mobile Commerce (M-Commerce): attributes, services, challenges

1.2 Exam Strategy Tips

- Learn all definitions precisely — exams often test exact meanings
- Know ALL e-business classification types (B2B, B2C, C2C, etc.) with examples
- Be able to draw/describe the B2B, B2C, and C2C flow diagrams
- Memorize the 6 Key Security Concerns (Privacy, Authenticity, Integrity, Non-repudiation, Access Control, Availability)
- Understand the difference between e-commerce and e-business clearly
- Know the 5 Attributes of M-Commerce (Ubiquity, Convenience, Interactivity, Personalization, Localization)
- Practice explaining each concept in 2-3 sentences

PART 2: TOPIC SUMMARIES

2.1 What is E-Business?

E-business (electronic business) is the use of Internet-based technologies to conduct all business operations — not just buying and selling, but also customer service and collaboration with partners.

Key Formula:

$$\text{e-business} = \text{Web} + \text{IT (Reliability, Security \& Availability)}$$

Two types of e-business structure:

- Pure Play – operates ONLINE ONLY (e.g., Amazon in early years)
- Brick and Click – operates both ONLINE AND OFFLINE (e.g., most modern retailers)

2.2 E-Business Classification Models

E-business is classified based on who the provider and consumer are:

Model	Full Name	Description
B2B	Business-to-Business	Transactions between companies (e.g., manufacturer to wholesaler)
B2C	Business-to-Consumer	Business sells directly to individual customers via website
C2C	Consumer-to-Consumer	Consumers sell to other consumers (e.g., eBay, Jumia classifieds)
C2B	Consumer-to-Business	Consumer sells product/service to a business online
B2G	Business-to-Government	Businesses transact with government agencies (variant of B2B)
G2B	Government-to-Business	Government interacts electronically with business enterprises
G2C	Government-to-Consumer	Government deals with citizens electronically
G2G	Government-to-Government	Inter-government electronic transactions
C2G	Consumer-to-Government	Citizens interact electronically with government
B2E	Business-to-Employee	Businesses deliver services/info to their own employees online

2.3 E-Business Security

Key Security Concerns:

- Privacy and Confidentiality – protecting user data from unauthorized access
- Authenticity – verifying the identity of users and systems
- Data Integrity – ensuring data is not altered in transit or storage
- Non-repudiation – a party cannot deny having sent/signed a communication
- Access Control – restricting who can access what data or systems

- Availability – ensuring systems are accessible when needed

Common Security Measures:

- Physical security of servers and hardware
- Secure data storage systems
- Data transmission protection and application development standards
- System administration controls

Security Solutions:

- Access and data integrity controls
- Encryption (e.g., 128-bit SSL)
- Digital certificates
- Digital signatures

2.4 E-Marketing

E-marketing (Electronic Marketing / Internet Marketing / Online Marketing) refers to the application of marketing principles via electronic media, especially the Internet.

Features of E-marketing:

- Allows global marketing from anywhere
- Less expensive than traditional marketing
- Makes marketing easier and more accessible
- Increases choice of products, services, and sellers
- Vast availability of information

Advantages of E-marketing:

- Inexpensive — reaches wide audiences at low cost
- Convenient research and purchase capability
- Pay-per-impression, pay-per-click, pay-per-action models
- 24/7 marketing — no time restrictions

E-Branding:

E-branding is the digital representation of a company's values, attitudes, vision, mission, personality, and appearance projected online. Its purposes are:

- Build direct relationships between customers and brand owners
- Strengthen brand identity
- Deliver consistent brand experience

Social media is the most powerful tool for promoting an e-brand.

2.5 E-Commerce

E-commerce is the buying and selling of products or services over electronic systems such as the Internet. It is a subset of e-business.

Key enabling technologies:

- EDI (Electronic Data Interchange) – structured transmission of business data between organizations electronically
- EFT (Electronic Funds Transfer) – electronic exchange of money between accounts

Advantages of E-Commerce:

- Faster buying/selling, easy product discovery
- 24/7 availability
- Lower operational costs, better service quality
- Easy to start and manage; no physical store required
- Customers can compare products across providers easily

Disadvantages of E-Commerce:

- No guarantee of product quality
- Vulnerable to hackers — sites, payment gateways always at risk
- No physical inspection of goods before purchase

E-Commerce Threats:

- Hackers stealing customer data or disrupting the site
- Stolen servers containing customer information
- Imposters creating mirror sites to steal customer money
- Hidden active content attacking the ecommerce system
- Disgruntled employees disrupting operations
- Competitor sabotage or data theft

2.6 Key Differences: E-Commerce vs. E-Business

Aspect	E-Commerce	E-Business
Meaning	Trading goods/services over the internet	All business activities conducted electronically
Scope	Narrow (commercial transactions)	Wide (all business operations)
Money	Always involves monetary transactions	May or may not involve money
Approach	Extroverted (outward-facing)	Ambiverted (internal + external)
Network	Uses the Internet only	Uses Internet, Intranet & Extranet
Requires	A website	Website + CRM + ERP + more
Relationship	Part of e-business	Broader concept; includes e-commerce

2.7 Mobile Commerce (M-Commerce)

M-Commerce is the buying and selling of goods and services through wireless handheld devices (mobile phones, PDAs, smartphones). It is essentially: M-Commerce = E-Commerce + Wireless Web.

5 Key Attributes of M-Commerce:

- Ubiquity – easier real-time information access from anywhere
- Convenience – mobile devices store data and connect to Internet/intranet/extranet
- Instant Connectivity – quick connection to Internet, databases, other devices
- Personalization – information tailored for individual consumers
- Localization – services matched to the user's current geographic location

M-Commerce Services & Applications:

- Mobile ticketing, content purchase & delivery, information services
- Mobile banking, mobile browsing, mobile purchase
- Mobile marketing and advertising
- Entertainment, communications, and transactions

M-Commerce Challenges & Limitations:

- Slow data transfer on 2G networks, raising customer costs
- Most websites not optimized for small mobile screens
- Tiny, awkward keyboards on cell phones
- Lack of universal standards
- Device constraints: weak processors, limited memory, poor resolutions

Protocols Used in M-Commerce:

- GSM (Global System for Mobile Communication) – uses TDMA technology
- 3G technologies: EDGE, UMTS (Universal Mobile Telecommunications System), HSPA
- Wireless Networks: Bluetooth, Wi-Fi
- SMS and MMS messaging technologies

PART 3: PRACTICE QUESTIONS & ANSWERS

These questions simulate the types of questions likely to appear in your final exam. Study each question and the corresponding answer carefully.

Section A: Definitions & Concepts

Q1. Define e-business and explain how it differs from traditional business.

Answer: E-business is the conducting of business on the Internet, using Internet-based technologies to transform key business processes both internally and externally. Unlike traditional business which relies on physical presence and face-to-face transactions, e-business leverages the web and IT infrastructure (data, applications, core processes) to perform all daily operations electronically, including customer service and partner collaboration.

Q2. What is a business model in the context of e-business?

Answer: A business model is defined as the organization of product, service, and information flows, and the source of revenues and benefits for both suppliers and customers. When organizations go online, they choose an e-business model that best aligns with their goals.

Q3. Distinguish between 'Pure Play' and 'Brick and Click' e-business.

Answer: Pure Play refers to an e-business that operates exclusively online with no physical presence (e.g., an online-only store). Brick and Click refers to a business that operates both online and offline — maintaining both a physical store and an online presence (e.g., a bank with branches and an online banking portal).

Q4. What is e-marketing? How does it differ from traditional marketing?

Answer: E-marketing (electronic marketing) refers to the application of marketing principles and techniques via electronic media, especially the Internet. Unlike traditional marketing (print, TV, radio) which is one-directional and more expensive, e-marketing is interactive, less costly, allows global reach, and operates 24/7. It enables pay-per-click/impression/action pricing models.

Q5. Define e-branding and state its purpose.

Answer: E-branding refers to the sum total of a company's values, attitudes, vision, mission, personality, and appearance projected online. Its purpose is to: (1) establish a direct relationship between customers and brand owners, (2) strengthen brand identity, and (3) deliver consistent brand experience. A brand tells who you are and what you stand for in the digital world.

Q6. What is e-commerce? How is it related to e-business?

Answer: E-commerce is the buying and selling of products or services over electronic systems such as the Internet and other computer networks. It is a major component of e-business. While e-commerce is limited to commercial (monetary) transactions, e-business is broader and encompasses all business activities conducted electronically, including internal processes, customer service, and partner collaboration.

Q7. Define Mobile Commerce (M-Commerce) and give its key formula.

Answer: M-Commerce is the buying and selling of goods and services through wireless handheld devices, and the use of mobile devices to communicate, transact, and entertain via connection to public and private networks. Its key formula is: $M\text{-Commerce} = E\text{-Commerce} + \text{Wireless Web}$. It enables transactions using mobile phones, PDAs, and smartphones.

Section B: Models and Classifications

Q8. List and briefly explain ALL e-business classification models.

Answer: There are 9 models classified by provider and consumer: (1) B2B – transactions between businesses; (2) B2C – business sells directly to consumers; (3) C2C – consumer sells to consumer; (4) C2B – consumer sells to business; (5) B2G – business transacts with government; (6) G2B – government interacts with businesses; (7) G2C – government deals with citizens electronically; (8) G2G – inter-government transactions; (9) B2E – business delivers services to employees.

Q9. Explain how the B2B e-business model works with a clear example.

Answer: In the B2B model, a website sells products to an intermediate buyer (not the final customer). For example, a wholesaler places an order on a manufacturer's website. The manufacturer processes the order and ships to the wholesaler. The wholesaler then sells the end product to final customers at retail outlets. Transactions occur between companies rather than between a company and individual consumers.

Q10. Describe how the B2C e-business model operates.

Answer: In the B2C model, a website sells products directly to individual customers. A customer visits the website, views products, selects and orders one. The website sends an email notification to the business organization, which then dispatches the product/goods directly to the customer. There is no intermediate business — the transaction is directly between the company and the end consumer.

Q11. How does the C2C e-business model work? Describe the three types of C2C communities.

Answer: In C2C, a website helps consumers sell assets (property, cars, motorcycles) to other consumers by publishing their listings. Consumer 1 places an advertisement; Consumer 2 views it and buys. The website may or may not charge a fee. Three C2C communities are: (1) Communities of Interest – people sharing common interests; (2) Communities of Relations – people with existing relationships; (3) Communities of Fantasy – virtual worlds and role-playing communities.

Q12. Explain the B2G e-business model.

Answer: The B2G (Business-to-Government) model is a variant of B2B. These websites are used by governments to trade and exchange information with various business organizations. They are accredited by the government and provide a platform for businesses to submit application forms and engage in procurement with the government. The flow is: Business Organization → Website → Government.

Section C: Security

Q13. List and explain the SIX key security concerns within e-business.

Answer: (1) Privacy and Confidentiality – protecting user data from unauthorized disclosure; (2) Authenticity – verifying the true identity of communicating parties; (3) Data Integrity – ensuring data is accurate, consistent, and unaltered throughout its lifecycle; (4) Non-repudiation – ensuring a party cannot deny sending a message or signing a document; (5) Access Control – restricting system access to authorized users only; (6) Availability – ensuring systems and data are accessible whenever needed.

Q14. What is non-repudiation? Why is it important in e-business?

Answer: Non-repudiation is the assurance that someone cannot deny something — specifically, it ensures a party to a contract or communication cannot deny the authenticity of their signature on a document or the sending of a message they originated. It is crucial in e-business for legal enforceability of contracts, payment authorizations, and order confirmations conducted electronically.

Q15. What are the four security solutions available for e-business systems?

Answer: The four security solutions are: (1) Access and Data Integrity controls — restricting who can read/modify data; (2) Encryption — encoding data using technologies like 128-bit SSL so only authorized parties can read it; (3) Digital Certificates — electronic documents that verify the identity of an entity online; (4) Digital Signatures — electronic equivalents of handwritten signatures for authenticating digital messages.

Q16. Name at least FIVE e-commerce threats.

Answer: (1) Hackers attempting to steal customer information or disrupt the site; (2) Physical theft of a server containing customer information; (3) Imposters creating mirror sites to steal customer money; (4) Authorized users downloading hidden active content that attacks the system; (5) A disaffected employee disrupting the ecommerce system; (6) Competitors or ex-employees accessing the site to cause disruption or steal data.

Section D: E-Commerce vs. E-Business & EDI/EFT

Q17. Compare and contrast e-commerce and e-business using at least FIVE dimensions.

Answer: (1) Meaning: E-commerce is online trading of goods/services; e-business is all business activities conducted electronically. (2) Scope: E-commerce is narrow; e-business is wide. (3) Money: E-commerce always involves monetary transactions; e-business may or may not. (4) Approach: E-commerce is extroverted (customer/supplier-facing); e-business is ambiverted (internal + external). (5) Network: E-commerce uses only the Internet; e-business uses Internet, intranet, and extranet. (6) Requirements: E-commerce needs just a website; e-business needs website + CRM + ERP.

Q18. Explain EDI and EFT and their role in e-commerce.

Answer: EDI (Electronic Data Interchange) is the structured transmission of data between organizations by electronic means — it transfers documents or business data from one computer system to another. EFT (Electronic Funds Transfer) is the electronic exchange or transfer of money from one account to another. Both technologies facilitated the original identification of e-commerce as a concept, enabling paperless business transactions and financial settlements between trading partners.

Section E: E-Marketing & M-Commerce

Q19. What are the FIVE features/advantages of e-marketing over traditional marketing?

Answer: (1) Global reach — internet marketing allows worldwide marketing facility; (2) Less expensive — far cheaper than print, TV, or radio campaigns; (3) Ease — makes marketing simpler to execute and manage; (4) More choice — increases available products, services, and sellers for consumers; (5) Vast information — provides enormous amounts of data for targeted campaigns. Additionally, e-marketing supports interactive advertising and 24/7 operation, unlike one-directional traditional marketing.

Q20. What are the FIVE attributes of M-Commerce? Explain each.

Answer: (1) Ubiquity — enables easier real-time information access from any location; (2) Convenience — devices store data and connect to Internet, intranet, and extranet; (3) Instant Connectivity — provides quick, easy connection to the Internet, intranets, databases, and other mobile devices; (4) Personalization — information and services are tailored and prepared for individual consumers; (5) Localization — the system knows where the user is located at any given time and matches services accordingly.

Q21. Describe THREE challenges/limitations of M-Commerce.

Answer: (1) Slow data transfer on second-generation (2G) cellular networks results in higher costs to customers; (2) Most web sites are not designed specifically for small wireless devices, making browsing difficult; (3) Keyboards and screens on cell phones are small and awkward to use. Additional limitations include: lack of industry standards, device constraints (weak processors, limited memory, tiny screens, poor resolution), and security vulnerabilities.

Q22. List FOUR M-Commerce services and applications.

Answer: M-Commerce offers: (1) Mobile ticketing — purchasing transport or event tickets via mobile; (2) Mobile banking — checking balances, making transfers, paying bills; (3) Mobile purchase — buying goods from online stores via smartphone; (4) Mobile marketing and advertising — targeted ads pushed to mobile devices. Other services include mobile browsing, content purchase and delivery, information services, and mobile payments.

Section F: Higher-Order / Application Questions

Q23. A company sells raw materials to manufacturers who then sell finished goods to retail shops. Which e-business model best describes each relationship? Justify your answer.

Answer: The relationship between the raw material supplier and the manufacturer is B2B (Business-to-Business), because both parties are businesses and transactions occur between companies with no direct end-consumer involvement. The relationship between the manufacturer and retail shops is also B2B for the same reason. If the retail shops eventually sell to individual customers (end consumers) through a website, that final link would be B2C (Business-to-Consumer). The entire chain illustrates a multi-layer B2B system culminating in B2C.

Q24. Explain why data integrity and non-repudiation are especially critical in B2B e-business transactions.

Answer: In B2B transactions, large sums of money, bulk orders, and legally binding contracts are exchanged electronically. Data integrity ensures that purchase orders, invoices, and contracts are not altered in transit — preventing costly errors or fraud. Non-repudiation is critical because it legally binds both parties to their commitments: a supplier cannot deny having received an order, and a buyer cannot deny having placed one. Without these two properties, B2B relationships would collapse due to disputes and lack of trust in electronic records.

Q25. Compare the infrastructure requirements of e-commerce and e-business. Why does e-business require more?

Answer: E-commerce requires only a website to conduct online buying/selling and uses only the Internet as its network. E-business, being broader, requires a website PLUS Customer Relationship Management (CRM) systems, Enterprise Resource Planning (ERP) systems, and other software packages. It also uses the Internet, intranet (internal network), and extranet (partner-accessible network). This is because e-business encompasses not just sales transactions but all business processes — HR, supply chain, finance, customer management — which demand integrated enterprise-level systems.

Q26. IremboGov is a Rwandan e-government platform where citizens apply for driving licenses online. Which e-business model does this represent? Explain the flow.

Answer: IremboGov represents the G2C (Government-to-Consumer/Citizen) model, where the government deals with citizens electronically. The flow is: the government provides an accredited website (IremboGov) → the citizen (consumer) accesses the portal → submits application forms → receives services (e.g., driving license). There is also a C2G (Consumer-to-Government) dimension, as citizens initiate transactions with the government. If businesses use the platform for regulatory compliance, it would also include B2G and G2B elements.

PART 4: QUICK REFERENCE CHEAT SHEET

Key Definitions at a Glance

Term	One-Line Definition
E-Business	Using internet tech to transform all business processes (Web + IT)
E-Commerce	Buying and selling goods/services over electronic systems (internet)
E-Marketing	Applying marketing principles via electronic/internet media
E-Branding	Company's digital identity, values, and appearance projected online
M-Commerce	E-Commerce conducted via wireless handheld mobile devices
B2B	Transactions between businesses (e.g., manufacturer to wholesaler)
B2C	Business sells directly to individual end consumers online
C2C	Consumers sell to other consumers via an intermediary website
B2G	Businesses transact electronically with government (variant of B2B)
G2C	Government delivers services to citizens electronically
EDI	Structured electronic data exchange between organizations
EFT	Electronic transfer of money between accounts
SSL	Secure Sockets Layer — encryption technology for safe web transactions
Pure Play	E-business that exists and operates online only
Brick & Click	Business operating both online and in physical locations
Non-repudiation	Assurance a party cannot deny a message or signature they originated
POS	Point-Of-Sale software for processing sales transactions electronically
GSM	Global System for Mobile Communication — digital mobile telephony standard

The 6 E-Business Security Concerns (Memorization Aid: PAINAA)

P – Privacy & Confidentiality A – Authenticity I – Integrity N – Non-repudiation A – Access Control A – Availability

5 M-Commerce Attributes (Memorization Aid: UCIPL)

U – Ubiquity C – Convenience I – Instant Connectivity P – Personalization L – Localization

E-Commerce vs E-Business: The 3-Point Summary

- E-Commerce = SUBSET of E-Business (commercial transactions only)
- E-Business = BROADER (all operations: HR, finance, supply chain, customer service + e-commerce)
- E-Commerce needs a website; E-Business needs website + CRM + ERP + intranet + extranet